

FitBuddy – AI Fitness Plan Generator

3. Project Design Phase

Architecture Design

FitBuddy follows a modular FastAPI-based architecture consisting of a frontend, backend, AI layer, and database.

Frontend Design

HTML, CSS, and Jinja2 templates are used for user input, result display, feedback, and administrator views.

Backend Design

FastAPI handles routing, form processing, validation, Gemini API calls, database operations, and template rendering.

AI Design

Gemini Pro is used for structured workout generation and feedback-based plan updates. Gemini Flash is used for fast nutrition/recovery tips.

Database Design

SQLite with SQLAlchemy stores user information and generated plans. Original and updated plans are preserved.

Main Pages

index.html for input; result.html for the workout plan, nutrition tip, and feedback; all_users.html for the administrator view.

Main Routes

/, /generate-workout, /submit-feedback, and /view-all-users.

Project: FitBuddy – AI Fitness Plan Generator using Gemini Models.